EXPERIMENT 1.2

3D Modelling of a Six-Hole Flange Plate

1. Aim

To create a two-dimensional profile of the given flange plate using the supplied dimensions, convert the profile into a three-dimensional solid by using the **Extrude** command, and prepare the required drawing views.

2. Requirements

Hardware	Computer or laptop
Software	CAD software with 2D sketching, trimming and 3D extrusion tools
Commands / tools	Line, circle, arc, trim, dimension, sketch and extrude
Reference	Given dimensioned sketch and model screenshots

3. Procedure

Step 1 – Note the dimensions: Before drawing the profile, refer to the dimensioned sketch. The visible callouts are **R21, Ø30, Ø32, R40, R32, R21 and R20** (as marked in the supplied drawing). Use the same units and scale as the CAD drawing. The extrusion thickness is not specified in the supplied dimension view, so use the thickness assigned by the experiment or instructor.

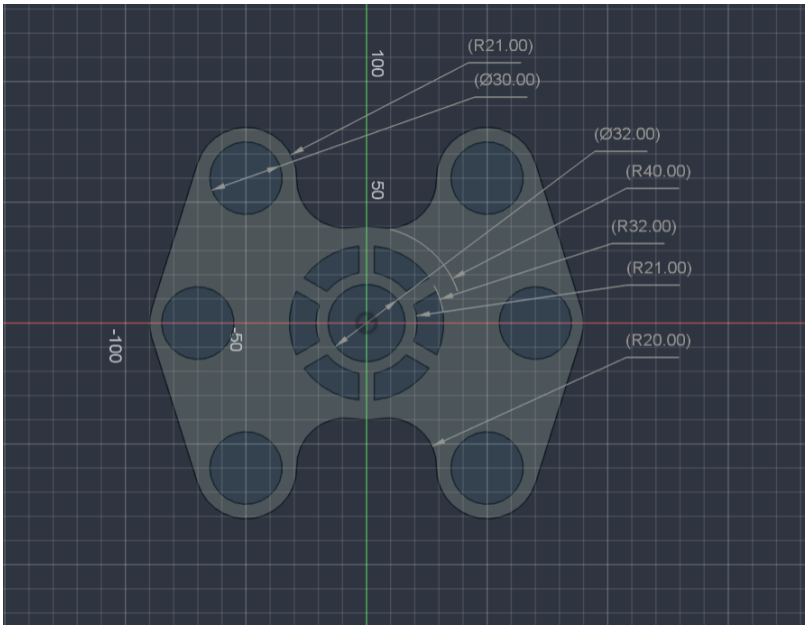


Figure 1. Dimensioned sketch of the flange plate.

Step 2 – Draw the 2D profile: Create the outer boundary symmetrically about the centre axes. Form the rounded upper and lower lobes and connect them with the side edges. Add the six equally arranged outer circular holes. At the centre, draw the circular opening and create the six separated curved slots around it, matching the reference sketch.

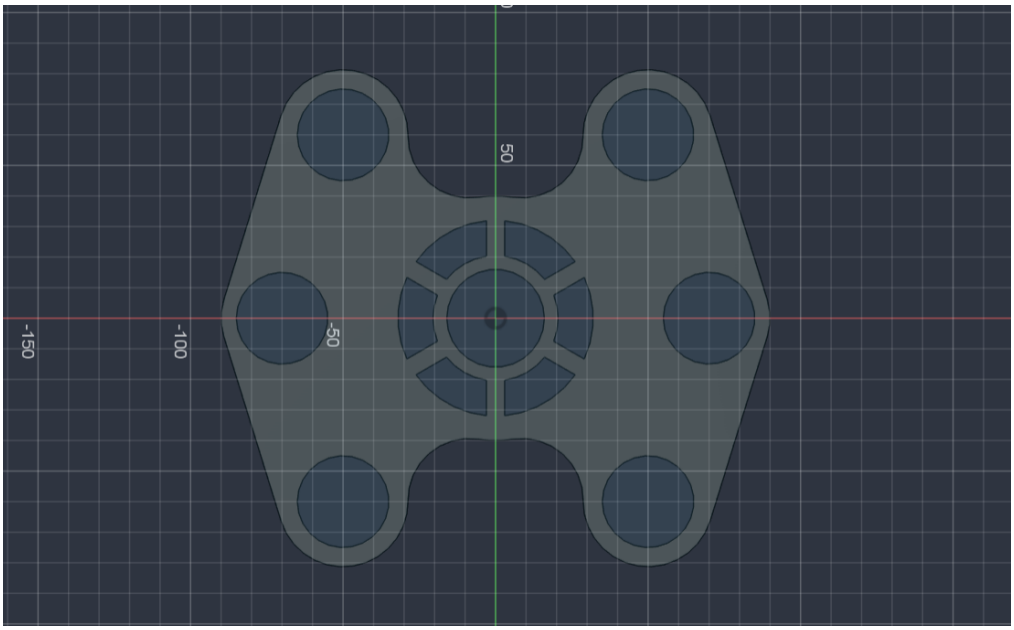


Figure 2. Top view of the completed 2D profile.

3. Procedure (continued)

Step 3 – Check the sketch: Verify that the profile and all openings match the reference. Ensure the outer boundary is a closed profile wherever a solid region is intended, and check that the six outer holes and central cut-outs are positioned correctly.

Step 4 – Extrude: Select the closed plate profile and apply the **Extrude** command. Enter the required thickness/depth from the experiment instructions, then confirm the operation to obtain the 3D flange plate. The image below shows the model after extrusion.

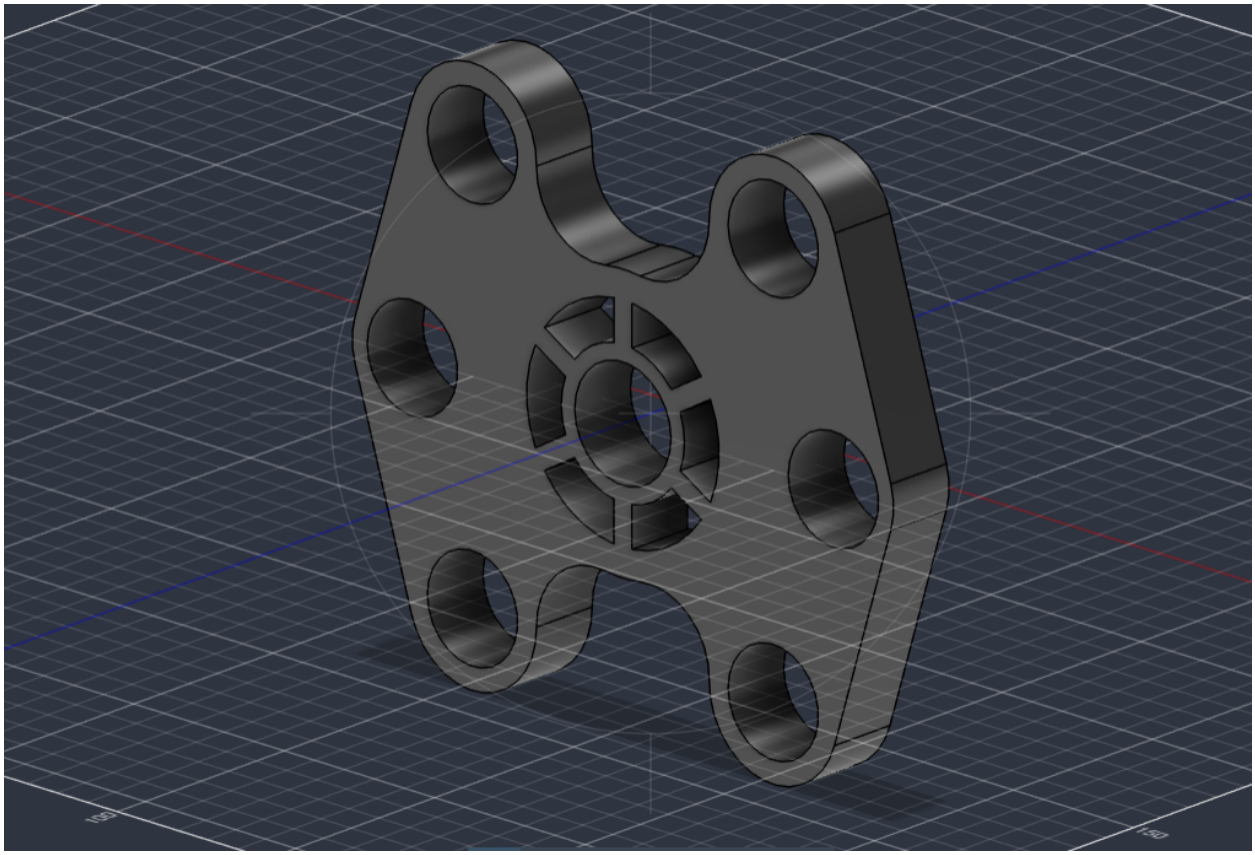


Figure 3. Isometric view of the flange plate after extrusion.

4. Output

The final output is the completed CAD drawing sheet shown below. It contains the orthographic views and isometric views of the model, arranged on the drawing sheet.

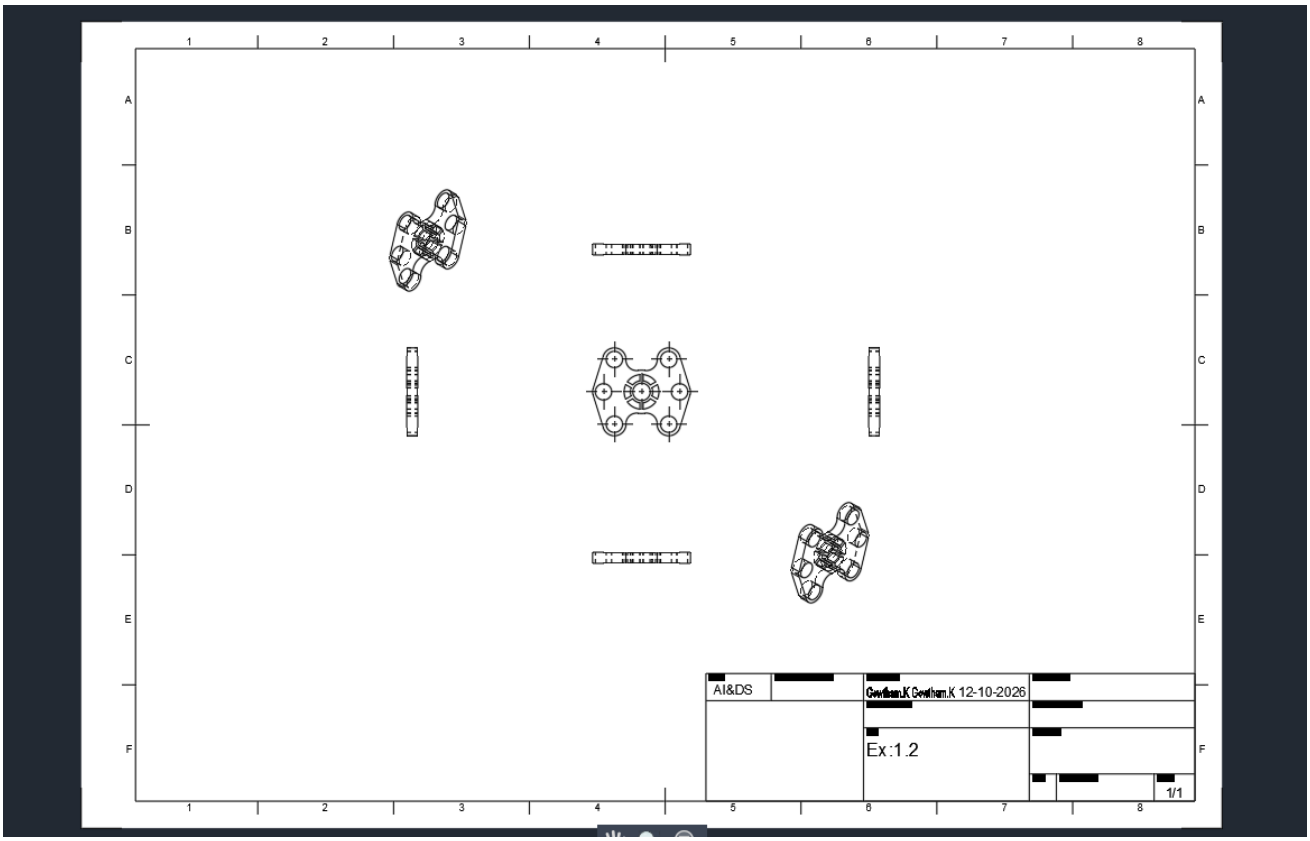


Figure 4. Final drawing sheet (last supplied picture).

5. Result

The flange plate was modelled successfully by creating the dimension-based 2D profile and extruding it into a 3D solid. The required holes, central opening and curved slots are represented, and the final drawing views were arranged on the sheet.

Note: The supplied dimensioned sketch does not show an extrusion depth. The actual depth should be entered according to the value used in the CAD model.